STAT 4714 - Midterm 1 Study Guide

February 2026

Problem 1

In a manufacturing process, 7% of microchips are defective. A testing device correctly identifies a defective chip with probability 0.88. The device incorrectly classifies a good chip as defective with probability 0.06.

1. Find the probability that a chip is classified as defective.
2. Given that a chip is classified as defective, find the probability that it is actually defective.

$P(D) = 0.07 \rightarrow P(G) = 0.93$

$P(\text{defective} \mid D) = 0.88$

$P(\text{defective} \mid G) = 0.06$

1) $P(\text{defective}) = P(\text{defective} \mid D) \cdot P(D) + P(\text{defective} \mid G) \cdot P(G) = (0.88)(0.07) + (0.06)(0.93)$
$= 0.1174$

2) $P(D \mid \text{defective}) = \dfrac{P(\text{defective} \mid D) \cdot P(D)}{P(\text{defective})} = \dfrac{(0.88)(0.07)}{0.1174}$

$= 0.525$

Problem 2

The joint density of two random variables X and Y is

$$f(x, y) = 6x, \quad 0 < x < 1,\ 0 < y < 1 - x.$$

1. Find the marginal density of X.
2. Find $P(X < 0.5)$.
3. Are X and Y independent?

Integrate opposite

1) $f(x,y) = 6x \rightarrow f_x(x) = \int_0^{1-x} 6x \, dy$

$= 6\left[xy\right]_0^{1-x}$

$= 6\left[\left(x(1-x)\right) - \left(x(0)\right)\right]$

$= 6\left[x - x^2\right]$

$= 6x(1-x), \quad 0 < x < 1$

2) use Marginal $\rightarrow P(X < 0.5) = \int_0^{0.5} 6x(1-x)\, dx$

$= 6\int_0^{0.5} x\,dx - 6\int_0^{0.5} x^2\,dx$

$= 6\left[\frac{x^2}{2}\right]_0^{0.5} - 6\left[\frac{x^3}{3}\right]_0^{0.5}$

$= 6\left[\left(\frac{0.5^2}{2}\right) - \left(\frac{0^2}{2}\right)\right] - 6\left[\left(\frac{0.5^3}{3}\right) - \left(\frac{0^3}{3}\right)\right]$

$= \cancel{6}^{3}\left(\frac{0.25}{\cancel{2}}\right) - \cancel{6}^{2}\left(\frac{0.125}{\cancel{3}}\right)$

$= 0.75 - 0.25 = 0.5$

3) $f(x,y) = f_x(x)\, f_y(y) \rightarrow$ Independent

Have $f_x(x)$, $f_y(y) = \int_0^{1-y} 6x \, dx$ $\nearrow$ Integrate opposite (since y bound contains x, use y bounds but in terms of x: $0 < y < 1-x$

$\downarrow$

$x < 1 - y$

$\downarrow$

$0 < x < 1-y$

$= 6\left[\frac{x^2}{2}\right]_0^{1-y} \rightarrow \cancel{6}^{3}\left[\frac{(1-y)^2}{\cancel{2}} - \frac{0^2}{2}\right] = 3(1-y)^2$

$6x(1-x) \cdot 3(1-y)^2 = 18x(1-x)(1-y)^2 \neq 6x \rightarrow$ $\boxed{\text{Not Independent}}$

Problem 3

The density function of a random variable is

$$f(x) = 5x^4, \quad 0 < x < 1.$$

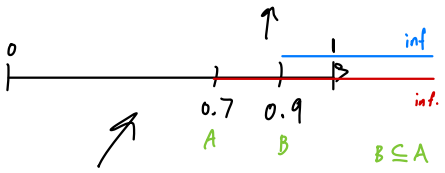
1. Find $P(X < 0.7)$.
2. Find $P(X > 0.9 \mid X > 0.7)$.

1) $P(x < 0.7) = \int_0^{0.7} 5x^4 \, dx = \cancel{5}\left[\frac{x^5}{\cancel{5}}\right]_0^{0.7}$

$= 0.7^5 - 0^5$

$= 0.168$

$P(x > 0.9)$ contains both



$B \subseteq A$

All vals of B are inside A

2) $P(x > 0.9 \mid x > 0.7) = \frac{P(A \cap B)}{P(B)} = P(x > 0.9 \cap x > 0.7)$

$\frac{P(x > 0.9)}{P(x > 0.7)} = \frac{0.410}{0.832} = \boxed{0.493}$

$P(x > 0.9) = \int_{0.9}^{1} 5x^4 = \cancel{5}\left[\frac{x^5}{\cancel{5}}\right]_{0.9}^{1} = (1)^5 - (0.9)^5$

$= 1 - 0.59 = 0.41$

$P(x > 0.7) = 1 - 0.168 = 0.832$

Problem 4

A system contains 4 independent components, each working with probability 0.95. The system functions if at least one component works. Find the probability that the system functions.

$P(\text{works}) = 0.95$ $\quad$ $P(\text{Broken}) = 0.05$ $\nearrow$ All 4 broken $= 0.05^4$

independent $\rightarrow$ $1 - 0.05^4 = 0.99$

Problem 5

The lifetime of a component has cumulative distribution function

$$F(x) = 1 - e^{-x/100}, \quad x > 0.$$

1. Find the probability density function.
2. Find the probability that lifetime exceeds 150.
3. Interpret $F(150)$ in plain English.

1) PDF is derivative of the CDF

$$F'(x) = f(x) = \frac{d}{dx}\left(1 - e^{-x/100}\right)$$

$$= \frac{d}{dx}(1) - \frac{d}{dx}\left(e^{-x/100}\right)$$

$$= 0 - \left(-\frac{1}{100} e^{-x/100}\right) = \frac{e^{-x/100}}{100}$$

2) $P(X > 150) = 1 - F(150) \longrightarrow$ sub into CDF

$$= 1 - \left(1 - e^{-150/100}\right) \quad 1 - e^{-3/2}$$

$$= 1 - 1 + e^{-3/2} = \boxed{e^{-3/2}}$$

3) $F(150)$ is the probability that the lifetime of the component fails at 150 units of time.

Problem 6

The joint density for random variables X and Y is

$$f(x, y) = 4xy, \quad 0 < x < 1,\ 0 < y < 1.$$

Find

1. $P(X < 0.5 \mid Y = 0.5)$.
2. σ_{XY}.

conditional distribution $\rightarrow \dfrac{f_{XY}(x,y)}{f_Y(y)}$

1) $P(X < 0.5 \mid Y = 0.5)$

We know $Y = 0.5 = 0$ so ... $f_Y(y) = \int_0^1 4xy\, dx = 4y\left[\frac{x^2}{2}\right]_0^1$

Marginal

$= \cancel{4}y \cdot \frac{1}{\cancel{2}} = 2y, \quad 0 < y < 1$

$f_{X|Y}(x|y) = \dfrac{4xy}{2y} = 2x, \quad 0 < x < 1$

$P(X < 0.5 \mid Y = 0.5) = \int_0^{0.5} 2x\, dx = \cancel{2}\left[\frac{x^2}{\cancel{2}}\right]_0^{0.5} = \boxed{0.25}$

2) σ_{XY} = Covariance $= E[XY] - E[X]E[Y]$

First ... $E[X] = \int x\, f_X(x)\, dx$

Marginal

$f_X(x) = \int_0^1 4xy\, dy \quad 0 < x < 1$

$= 4x\left[\frac{1}{2}\right] = 2x$

$= 2\int_0^1 x^2\, dx = 2\left[\frac{x^3}{3}\right]$

$= 2\left[\frac{1}{3}\right] = \frac{2}{3}$

$E[Y]$ = by symmetry

$E[XY] = \iint xy\, f_{XY}(x,y)\, dx\, dy = \int_0^1\int_0^1 4x^2y^2\, dx\, dy = \int_0^1 4y^2\left[\frac{x^3}{3}\right]_0^1 dy = \int_0^1 \frac{4y^2}{3}\, dy$

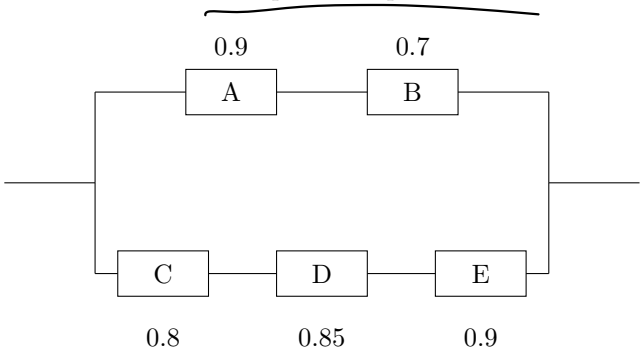
$= \frac{4}{3}\left[\frac{y^3}{3}\right]_0^1$

$= \frac{4}{3}\cdot\frac{1}{3} = \frac{4}{9}$

$\sigma_{XY} = \frac{4}{9} - \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) = \frac{4}{9} - \frac{4}{9} = \boxed{0}$

Problem 7

A circuit system is given below. Assume independent operation.



Find the probability that the system works.

AB → Series → $(0.9)(0.7) = 0.63$

CDE → Series → $(0.8)(0.85)(0.9) = 0.612$

at least 1 works ? → $1 - (1 - R_{Top})(1 - R_{Bottom})$

$= 1 - (1 - 0.63)(1 - 0.612)$

$= 1 - (0.37)(0.388)$

$= 1 - 0.14356$

$= 0.85644$

Problem 8

The density of a random variable X is

$$f(x) = \frac{1}{2}(2 - x), \quad 0 < x < 2.$$

Find

1. $P(X < 1)$.
2. $E(X)$.

1) $P(X<1) = \int_0^1 \frac{1}{2}(2-x)\,dx \rightarrow 1 - \frac{1}{2}x$

$= 1\int_0^1 dx - \frac{1}{2}\int_0^1 x\,dx$

$= [x]_0^1 - \frac{1}{2}\left[\frac{x^2}{2}\right]_0^1$

$= 1 - \frac{1}{4} = \boxed{\frac{3}{4}}$

2) $E[x] = \int x\, f(x)\,dx = \int_0^2 x\left(1 - \frac{1}{2}x\right)dx \rightarrow x - \frac{1}{2}x^2$, $0 < x < 2$

$= \int_0^2 x\,dx - \frac{1}{2}\int_0^2 x^2\,dx$

$= \left[\frac{x^2}{2}\right]_0^2 - \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2$

$= \frac{4}{2} - \frac{1}{2}\left(\frac{8}{3}\right)$

$= \frac{4}{2} - \frac{8}{6}$

$= \frac{12}{6} - \frac{8}{6} = \frac{4}{6} = \boxed{\frac{2}{3}}$

Problem 9

The lifetime of an electronic component is a random variable X with mean 1000 hours and standard deviation 200 hours. Provide a lower bound for the probability that the lifetime is between 600 and 1400 hours. Clearly state the inequality you use and interpret your result.

$N = 1000$ hrs

$\sigma = 200$ hours

$P(600 \leq x \leq 1400) = ?$

use Chebyshev's for finding lower bounds

$P(|x - N| < k\sigma) \geq 1 - \frac{1}{k^2}$ OR $\leq \frac{1}{k^2}$

$600 = \underset{\uparrow \atop N}{1000} - 400$ $\qquad$ $1400 = 1000 + 400$

$P(600 \leq x \leq 1400) = P(|x - 1000| \leq 400)$

$k = \frac{400}{200} = 2$

$P(|x - 1000| < 2\sigma) \geq 1 - \frac{1}{2^2}$

$= 1 - \frac{1}{4} = 3/4$

$\rightarrow P(600 \leq x \leq 1400) \geq 0.75$

This means at least 75% of electronic components will have a lifetime between 600 and 1400 hours.

Problem 10

Let X be a discrete random variable representing the number of defects found in a manufactured item. The probability mass function of X is given by

$$P(X = 0) = 0.3, \quad P(X = 1) = 0.4, \quad P(X = 2) = 0.3.$$

1. Find the mean (expected value) of X.
2. Find the variance of X.

1) $E[X] = \sum x \cdot f(x) = (0)(0.3) + (1)(0.4) + (2)(0.3)$

$= 0.4 + 0.6$

$= \boxed{1}$

2) $Var(X) = \sigma^2 = E(x^2) - \mu^2$

$(0^2)(0.3) + (1^2)(0.4) + (2^2)(0.3)$

$= 0.4 + 1.2$

$= 1.6$

$= 1.6 - 1^2 = \boxed{0.6}$

Problem 11

Consider the following probability density function for a random variable X:

$$f(x) = 2(1-x), \quad 0 < x < 1.$$

1. Find the probability that X exceeds 0.5.
2. Given that X exceeds 0.5, find the conditional probability that X is less than 0.75.

1) $P(X > 0.5) = ?$

$$\int_{0.5}^{1} (2-2x)\,dx = 2[x]_{0.5}^{1} - \cancel{2}\left[\frac{x^2}{\cancel{2}}\right]_{0.5}^{1}$$

$$= 1 - 0.75$$

$$= \boxed{0.25}$$

2) $P(X < 0.75 \mid X > 0.5) = ?$

$$\frac{P(0.5 < X < 0.75)}{P(X > 0.5)} = \frac{0.1875}{0.25} = \boxed{0.75}$$

$$\int_{0.5}^{0.75} (2-2x)\,dx = 2[x]_{0.5}^{0.75} - \cancel{2\left[\frac{x^2}{2}\right]_{0.5}^{0.75}}$$

$$= 0.5 - 0.3125$$

$$= 0.1875$$

Problem 12

The joint probability density function of two random variables X and Y is given by

$$f(x, y) = 2, \quad 0 < x < y < 1.$$

$x > 0$ $\quad x < y$

$y < 1$

1. Find the expected value $E[X]$.
2. Find the expected value $E[Y]$.

dependent always inside!

both x and Y are between 0 and 1

X is always less than Y!!

1) $E[X] = \iint x\, f(x,y)\, dx\, dy = \int_{y=0}^{1} \int_{x=0}^{y} 2x\, dx\, dy$

$= \cancel{2} \int_{y=0}^{1} \left[\frac{x^2}{\cancel{2}} \right]_0^y dy$

$= \int_{y=0}^{1} y^2\, dy = \left[\frac{y^3}{3} \right]_0^1 = \boxed{\frac{1}{3}}$

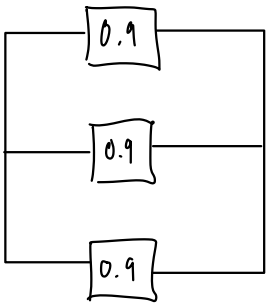
2) $E[Y] = \iint y\, f(x,y)\, dx\, dy = \int_{y=0}^{1} \int_{x=0}^{y} 2y\, dx\, dy$

$= 2 \int_{y=0}^{1} [xy]_0^y\, dy = 2 \int_{y=0}^{1} y^2\, dy$

$= 2 \left[\frac{y^3}{3} \right]_0^1 = \boxed{\frac{2}{3}}$

Problem 13

A system contains three independent components connected in parallel. Each component operates successfully with probability 0.9. The system operates successfully if at least one component operates. Find the probability that the system operates successfully.



$P(\text{succeeds}) = 0.9 \quad P(\text{fails}) = 1 - 0.9 = 0.1$

$\text{All works} = 1 - (0.1)^3 = \boxed{0.999}$

Problem 14

The lifetime of a certain electronic component is a random variable X with probability density function

$$f(x) = \frac{1}{4}e^{-x/4}, \quad x > 0.$$

1. Find the mean lifetime of the component. → Expected Value
2. Find the variance of the lifetime.

$$E[X] = \int_0^\infty x \cdot f(x) = \frac{1}{4}\int_0^\infty x e^{-x/4}\, dx$$

↑

need integration by parts...

we see it's an exponential distrobution → $\lambda e^{-\lambda x}$

$$\text{Mean} = \frac{1}{\lambda} = \frac{1}{\frac{1}{4}} = \boxed{4}$$

$$\text{Var} = \frac{1}{\lambda^2} = \frac{1}{\frac{1}{16}} = \boxed{16}$$

Problem 15

The joint probability density function of two random variables X and Y is

$$f(x, y) = 8, \quad 0 < x < y < 0.5.$$

$X > 0$ $\quad X < Y$

$Y < 0.5$

Find the correlation coefficient between X and Y.

$$\rho = \frac{Cov(X,Y)}{\sigma_X \ \sigma_Y} \longrightarrow Cov(X,Y) = E(XY) - E[X]\,E[Y]$$

$$= 0.0625 - (0.167)(0.333) = 0.006889$$

$$E[X] = \iint x \cdot 8 \, dx\, dy = 8 \int_{y=0}^{0.5} \int_{x=0}^{y} x \, dx\, dy$$

$$= \cancel{8}^{\,4} \int_{y=0}^{0.5} \left[\frac{x^2}{\cancel{2}} \right]_0^y dy$$

$$= 4 \left[\frac{y^3}{3} \right]_0^{0.5}$$

$$= \frac{4}{3} \left[0.5^3 \right] = 0.167$$

$$E[Y] = 0.5 - 0.167 = 0.333$$

$$E[XY] = \iint xy \, f(x,y) \, dx\, dy = 8 \int_{y=0}^{0.5} \int_{x=0}^{y} xy \, dx\, dy$$

$$= 8 \int_0^{.5} \left[\frac{x^2 y}{2} \right]_0^y dy = \cancel{8}^{\,4} \int_0^{0.5} \left(\frac{y^3}{\cancel{2}} \right) dy$$

$$= \cancel{4} \left[\frac{y^4}{\cancel{4}} \right]_0^{0.5}$$

$$= 0.0625$$

$$Var(X) = E[X^2] - [E(X)]^2 = 0.0417 - (0.167)^2 = \sqrt{0.0138} = 0.11747340$$

$$Var(Y) = E[Y^2] - [E(Y)]^2 = 0.125 - (0.333)^2 = \sqrt{0.0141}$$

Problem 15

The joint probability density function of two random variables X and Y is

$$f(x, y) = 8, \quad 0 < x < y < 0.5.$$

Find the correlation coefficient between X and Y.

$$E[X^2] = 8 \int_{y=0}^{0.5} \int_{x=0}^{y} x^2 \, dx \, dy$$

$$= 8 \int_0^{0.5} \left[\frac{x^3}{3} \right]_0^y dy = 8 \int_0^{0.5} \frac{y^3}{3} dy$$

$$= \frac{\cancel{8}^{2}}{3} \left[\frac{y^4}{\cancel{4}} \right]_0^{0.5} = \frac{2}{3}(0.0625)$$

$$= 0.0417$$

$$E[Y^2] = 8 \int_0^{0.5} \int_0^{y} y^2 \, dx \, dy$$

$$= 8 \int_0^{0.5} \left[xy^2 \right]_0^y dy = 8 \int_0^{0.5} y^3 \, dy$$

$$= \cancel{8}^{2} \left[\frac{y^4}{\cancel{4}} \right]_0^{0.5}$$

$$= 0.125$$

$$\rho = \frac{0.006889}{0.117473401^2} = 0.5$$